

Which Frames Matter? Frame Selection for Training Object Detectors on Driving Videos

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Abstract

Although training accurate object detectors requires large-scale annotated data, continuous driving video is dominated by near-duplicate frames that not only waste annotation budget but also negatively impact generalization. Existing frame selection methods either ignore temporal structure entirely or exploit it only locally, without enforcing global diversity. Additionally, current benchmarks lack the dense annotations required to properly evaluate frame selection. This paper addresses these gaps by introducing AVADiP-DFS, the first benchmark explicitly designed for evaluating frame selection for object detector training, comprising 160 driving video sequences (48,000 frames at 30 fps) with exhaustive bounding-box annotations and a standardized multi-budget evaluation protocol. We also propose SHIFT (Selecting High-Information Frames for Training), a training-free method that reduces training set redundancy by allocating candidates proportionally to visual change via cumulative SSIM variation, then refines them through greedy entropy maximization over pretrained embeddings, enforcing global diversity. Extensive experiments show that SHIFT surpasses full-dataset performance using only 1% of frames (+3.6 mAP) and outperforms all baselines by up to +5.6 mAP points using only one third of the data, demonstrating that effective frame selection simultaneously reduces annotation cost and improves detection accuracy.

1. Introduction

Object detection is a cornerstone of autonomous driving perception, enabling vehicles to perceive and react to pedestrians, cyclists, and surrounding traffic in real time [7]. While the continuously growing volume of dash-

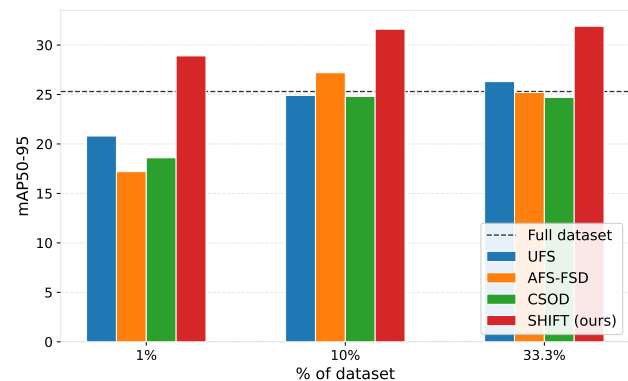


Figure 1. SHIFT consistently outperforms all competing methods across annotation budgets, surpassing full-dataset performance (dashed line) with only 1% of frames and reaching a gain of +6.6 mAP with one third of the data.

cam footage from ride-sharing platforms and advanced driver-assistance systems (ADAS) seems to offer a natural solution to the data scarcity problem, quantity alone does not address the annotation bottleneck: training accurate detectors still requires large-scale, carefully annotated datasets, which is expensive and time-consuming task.

Nonetheless, the vast majority of frames in a driving video are near-duplicates of their temporal neighbors: vehicles following a steady trajectory, red lights, and long highway stretches with little visual variation routinely produce very similar consecutive frames, typically with structural similarity index measure (SSIM) values above 0.95 [26].

Annotating all frames is not only inefficient, but also degrades detector performance, as redundant training examples increase generalization error without providing new visual evidence [20, 26]. This raises a practical question: given a fixed annotation budget, which frames should be selected to maximize detection performance?

Current approaches address parts of this problem but fall

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short of a complete solution. Coreset selection and data pruning methods aim to extract representative subsets from large image collections [4, 19]. Although recent efforts target object detection [14], these methods treat datasets as unordered pools of independent images, ignoring the temporal structure intrinsic to video and offering no mechanism to suppress the near-duplicate frames that dominate driving sequences.

Conversely, frame selection has been explored extensively in video understanding [11, 18, 22], where adaptive sampling and saliency strategies individually select informative frames for a single video-level prediction, scoring them in isolation without penalizing redundancy among selections. This is sufficient for a single prediction per video, but unsuitable for assembling a training set under a fixed annotation budget, where every redundant selection wastes an opportunity to introduce new object categories, scales, or scene conditions into the training distribution.

The only prior work that directly addresses frame selection for detector training is Yoon and Choi [26], who show that redundant frames are not merely unnecessary but detrimental, and that subsampling consistently improves performance. Their evaluation, however, relies on BDD100K MOTS [27] dataset, which provides annotations at only 5 fps against a 30 fps source, partially masking the redundancy structure of real driving video and restricting evaluation to a narrow range of annotation budgets. Fundamentally, the field lacks a benchmark with dense, per-frame annotations, as well as an evaluation protocol for assessing frame selection in detector training.

In this paper, we address this gap with two contributions. First, we introduce the dataset AVADiP Dashcam Frame Selection (AVADiP-DFS)¹, a benchmark explicitly designed for evaluating frame selection strategies for training object detectors on driving video. It comprises 160 video sequences of 10 seconds recorded from front and rear perspectives, totaling 48,000 frames at 30 fps with exhaustive per-frame bounding-box annotations across six object categories. The dense annotation coverage enables controlled evaluation across a wide range of annotation budgets, isolating the effect of frame selection from confounding factors such as annotation protocol or temporal subsampling.

Second, we propose **SHIFT** (Sparse High-Information Frames for Training), a training-free frame selection method that jointly addresses temporal non-redundancy and semantic diversity. At its core, SHIFT treats frame selection as an information maximization problem: each selected frame should contribute visual evidence that no other selected frame already provides. To this end, it concentrates candidates in the most visually dynamic regions of the video via a cumulative SSIM-based variation signal, then refines

this pool through greedy entropy maximization over pre-trained visual embeddings. The entire pipeline requires no ground-truth annotations or pretrained detector, making it a practical, detector-agnostic preprocessing step for any annotation workflow.

Extensive experiments on our AVADiP-DFS benchmark validate the approach. As shown in Figure 1, using only **1%** of frames, SHIFT already exceeds the mAP of a detector trained on the full dataset by **+3.6** points, confirming that the vast majority of frames in driving video are redundant for training. At its peak, SHIFT reaches **31.9** mAP using 33% of the data, outperforming the full-dataset reference (25.3 mAP) by **+6.6** points and competing methods at the same budget by **+5.6** points. Qualitative analysis shows that the method naturally allocates more frames to dynamic and transitional scenes and fewer to static or repetitive ones, aligning annotation budget with information content.

In a nutshell, in this paper we:

- introduce AVADiP-DFS, the first benchmark for evaluating frame selection for detector training on driving video, featuring 160 sequences with exhaustive 30 fps bounding-box annotations and a standardized multi-budget evaluation protocol.
- propose SHIFT, a training-free frame selection method that jointly addresses temporal non-redundancy and semantic diversity, requiring no detector training or labels during selection.
- demonstrate through extensive experiments that SHIFT surpasses full-dataset performance with only 1% of frames and outperforms all comparative methods by up to **+5.6** mAP points, with consistent gains across all evaluated annotation budgets.

2. Related Work

2.1. Data Selection

Selecting a compact yet representative subset of training data has been extensively studied in the context of image classification, with approaches ranging from geometry-based coverage methods [19, 25] and training-dynamics scoring [17, 23] to submodular optimization to balance representativeness and diversity [9, 24]. Extending these ideas to object detection is non-trivial, since each image may contain multiple objects of different categories and scales, and a single image-level score cannot capture this complexity. CSOD [14] addressed this by extracting per-object RoI features from ground-truth boxes, aggregating them into image-classwise representations, and selecting a subset via submodular optimization. While CSOD achieves substantial gains on Pascal VOC [5], it requires full bounding-box annotations for every candidate image prior to selection, limiting its applicability to scenarios where labels already exist. Critically, all these methods treat datasets as unordered collections of independent images, with no mecha-

¹Our code and dataset are publicly available at <https://github.com/viniavena/shift-datacv2026>.

nism to account for the temporal structure intrinsic to video.

2.2. Frame Selection in Videos

Frame selection has been studied across a wide range of video understanding tasks to reduce inference cost or annotation effort. For action recognition, learned sampling policies [11] and training-free strategies based on motion signals [28] select the most informative frames to process from long untrimmed videos. In video question answering, frames are selected conditioned on the input query [1, 22], making frame importance inherently input-dependent. In temporal action detection, ASL [18] reduces annotation cost through active learning, acquiring frames iteratively based on a partially trained model. While some of these methods are effective in their respective settings, all of them optimize for a single prediction per video and select frames in isolation, without penalizing redundancy among selections. This property, sufficient for video-level inference, is fundamentally unsuitable for assembling a diverse training set under a fixed annotation budget.

The most directly related work is Yoon and Choi [26], who studied how temporal redundancy affects instance segmentation. Their key finding is that subsampling to as little as 50% of frames improves performance on BDD100K MOTS [27], confirming that redundancy is not merely wasteful but harmful. Nonetheless, their variance-based scoring captures only local, consecutive-frame dissimilarity without enforcing global diversity. The authors themselves identified diversity and more adaptive per-video strategies as open directions.

Existing driving datasets further limit progress in this area. BDD100K MOTS annotates at only 5 fps from 30 fps source videos, nuScenes [2] at 2 fps from 12 fps, and BDD100K detection [27] labels a single keyframe per 40-second clip. While Waymo [21] and KITTI [6] provide dense annotations, their 10 fps capture rate still restricts the range of testable budgets. In all these cases, the most redundant frames have already been discarded before any selection method is applied. This means that no existing dataset exposes the full temporal redundancy of driving video or provides the dense per-frame annotations and standardized protocol needed to systematically benchmark frame selection for detector training. Our benchmark is designed to fill this gap.

3. AVADiP Dashcam Frame Selection

3.1. Dataset

We introduce AVADiP-DFS, a driving video dataset designed to support the evaluation of frame selection methods for object detector training under realistic temporal redundancy conditions. It comprises 160 driving video clips of 10 seconds each, captured at 30 fps and high resolution using front and rear onboard cameras. The sequences cover

a range of environmental conditions: road type, traffic density, time of day, and weather. Camera perspective is balanced between front and rear views.

All sequences were annotated manually by trained annotators, producing exhaustive per-frame bounding-box labels across six object categories: *Person*, *Motorcycle*, *Animal*, *Bicycle*, *Car*, and *Heavy Vehicle*. Every frame in every sequence was labeled, yielding a total of 48,000 annotated frames. Unlike existing driving datasets, which subsample frames prior to annotation, AVADiP-DFS preserves the full temporal redundancy of the source video, enabling controlled evaluation of frame selection methods across realistic annotation budgets.

3.2. Comparison with Existing Driving Datasets

Existing large-scale driving datasets cover a wide range of scenarios, spanning multiple geographical domains, weather conditions, and challenging detection contexts, such as poor illumination and occlusion [15]. While this curated diversity has enabled the development of models with better generalization capabilities, it often involves the exclusion of many redundant frames, resulting in sparsely annotated data, as discussed in Section 2.2. Table 1 summarizes this limitation across several datasets. BDD100K MOTS [27] and nuScenes [2] provide annotations for only one out of six frames, while BDD100K detection [27] labels a single keyframe for each 40-second video. Such sparsely annotated data poses a challenge for evaluating frame selection methods, as it limits the scope of possible selections to those smaller than the already filtered subset.

Table 1. Comparison of temporal density and frame selection support across driving datasets. BDD100K D and M denote the Detection and MOTS subsets, respectively. Annotation and Source rates are expressed in FPS.

Dataset	Source Rate	Annot. Rate	Dense Annot.	Selection Support
nuScenes [2]	12	2	No	No
BDD100K D [27]	30	-	No	No
BDD100K M [27]	30	5	No	Limited
Waymo [21]	10	10	Yes	Limited
KITTI [6]	10	10	Yes	Limited
AVADiP-DFS (ours)	30	30	Yes	Yes

The lack of full annotation coverage of existing driving datasets, as shown in the column “Dense Annotations” of Table 1, limits their applicability as a benchmark for the evaluation of frame selection methods, since these pre-sampling strategies effectively mask the highly redundant nature present in continuous driving video, restricting the scope of testable annotation budgets. Although datasets such as Waymo [21] and KITTI [6] offer densely annotated data, their lower capture rate produces similar restrictions.

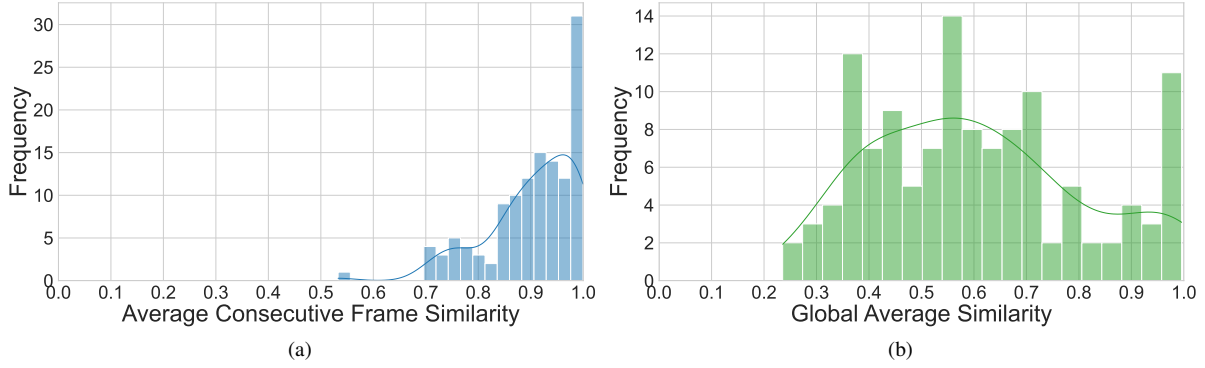


Figure 2. Average Consecutive Frame Similarity and Global Average Similarity measures to summarize the temporal structure of AVADiP-DFS. The ACFS distribution confirms high local similarity between consecutive frames, while the broad GAS distribution reflects diverse visual content across the video clip.

In contrast, AVADiP-DFS provides exhaustive, frame-level annotations at a 30 fps rate, allowing for the evaluation of frame selection methods under a more realistic scenario of highly redundant sequential data.

3.3. Redundancy Analysis

To characterize the temporal structure of AVADiP-DFS, we analyzed the redundancy present in its sequences through metrics derived from pairwise SSIM across all frames of each video. Average Consecutive Frame Similarity (ACFS) measures the average SSIM between consecutive frames, capturing local frame-to-frame resemblance. Global Average Similarity (GAS) measures the average SSIM across all frame pairs in a video, reflecting how visually homogeneous the video is as a whole.

Figure 2 reveals two complementary dimensions of redundancy characteristic of driving videos. The ACFS distribution is heavily skewed toward 1.0, confirming that consecutive frames are nearly identical. This is consistent with the findings of Yoon and Choi [26], and implies that training on all frames is both wasteful and harmful to detector generalization. GAS, however, exhibits a bimodal distribution, with one mode around 0.4–0.6 and a secondary peak near 1.0, revealing a dataset that spans from moderately dynamic sequences to nearly static ones. Crucially, a sequence can exhibit high consecutive frame similarity yet low global average similarity: locally, frames vary little between adjacent time steps, while globally the visual content changes substantially over time. This is precisely the regime where methods based on uniform sampling fail, selecting frames at fixed intervals without adapting to where informative content is concentrated.

To further analyze the temporal behavior of frame redundancy, we examine how the similarity across frames evolves as the time interval separating them increases. Figure 3 shows the average similarity as a function of the temporal lag, measured with both SSIM and the cosine distance between ResNet-50 feature embeddings. Each point corre-

sponds to the average similarity calculated across all frame pairs separated by the same time distance, aggregated over all videos in the dataset.

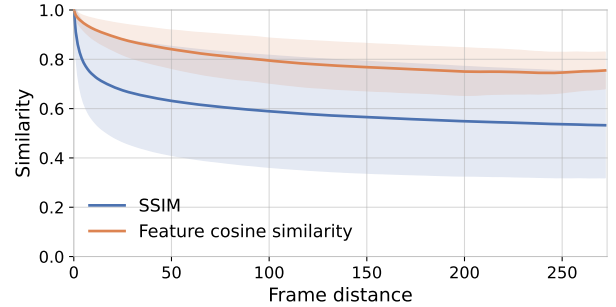


Figure 3. Average similarity between frames as a function of temporal distance. Curves represent the mean similarity across all videos in the dataset, while the shaded region indicates ± 1 standard deviation.

Both curves decay as the number of frames between the compared pairs increases, with a sharper decline at short temporal distances. Temporally close frames are nearly identical, yielding average similarity values approaching 1.0 for both metrics. As the temporal lag grows, similarity gradually decreases, indicating that visual changes accumulate over time. However, after an initial lag interval, the decay becomes smoother, suggesting that visual information often persists over the short driving video sequences present in the proposed dataset. While SSIM decreases more rapidly, likely due to its higher sensitivity to pixel-level changes such as motion blur, the cosine similarity between feature embeddings shows a slower decay over time. This suggests that, even though small visual details may vary throughout a video, relevant semantic information redundancy tends to persist over larger time intervals.

The disparity between the rate of change in pixel-level and semantic redundancy underscores the necessity for

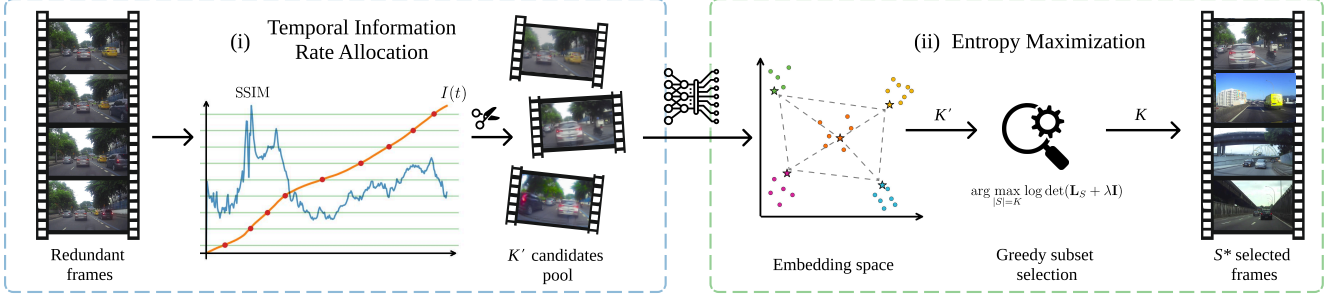


Figure 4. Overview of the SHIFT pipeline. (i) Temporal Information Rate Allocation samples the cumulative information signal $I(t)$ to extract candidate frames, densely covering dynamic segments while compressing static scenes. (ii) Entropy Maximization refines these candidates to the final budget K by maximizing the log-determinant in the embedding space, favoring visually complementary frames and reducing redundancy.

frame selection methods considering both dimensions of video temporal dynamics. Simple time-centric approaches, such as a uniform sampling over time, may be able to select a varied set of frames in terms of pixel-wise characteristics, but have no guarantees against semantic redundancy. In contrast, SHIFT employs a two-step pipeline to jointly optimize for local temporal redundancy mitigation and global semantic diversity.

3.4. Evaluation Protocol

We define an evaluation protocol for frame selection methods applied to object detector training on driving video. We split the 160 video sequences into 110 for training, 15 for validation, and 35 for testing at the video level, ensuring no sequence contributes frames to multiple splits. Frame selection methods operate exclusively on the training set: given a target annotation budget, each method selects a subset of frames, and a detector is trained on the corresponding annotations. Evaluation is performed on the complete test split at the original 30 fps, ensuring that reported metrics reflect generalization to the full temporal distribution.

Budget levels are defined to match the frame counts produced by uniform sampling at each target rate, establishing a direct correspondence between budget and the trivial baseline. We evaluate all methods at seven budget levels spanning 0.3 to 30 fps (1% to 100% of the full stream), where the full-dataset (100%) condition serves as a reference.

Performance is reported as COCO-style mAP, averaging precision across IoU thresholds from 0.5 to 0.95 in steps of 0.05, over the six object categories. All models use YOLO11X [8], initialized from COCO-pretrained weights and fine-tuned under identical training settings, so that the selected subset is the sole experimental variable.

4. Proposed Method: SHIFT

Selecting an informative subset of frames for detector training from driving videos is a constrained resource-allocation problem. As shown in Section 3, consecutive frames are highly redundant, making exhaustive training on all T

frames inefficient in practice. Given an annotation and training budget of K frames ($K \ll T$), the goal is to choose, independently for each video, a subset S^* that retains the most useful training information while avoiding near-duplicate samples. The final training set is formed by aggregating the selected subsets across all videos.

To address this, we propose **S**parse **H**igh-**I**nformation **F**rames for **T**raining (SHIFT), a method that formulates frame selection as an entropy-oriented optimization problem in feature space. Instead of searching over all T frames directly, which is intractable for long videos, SHIFT adopts an efficient temporal-diversity decomposition, as depicted in Figure 4. First, a temporal allocation step samples candidate frames according to the rate of visual change, prioritizing dynamic segments. Then, a feature-space diversity refinement step selects the final budget K frames by maximizing a log-determinant objective over the candidate embeddings, favoring subsets that cover complementary visual content while reducing redundancy.

4.1. Temporal Information Rate Allocation

To address with temporal redundancy in frame selection, SHIFT allocates the frame sampling budget proportionally to visual change rather than uniformly in time. For a video of T frames, we compute a variation score $v_t = 1 - \text{SSIM}(f_t, f_{t-1})$ at each frame t , and accumulate these scores into a monotonic signal $I(t) = \sum_{s=1}^t v_s$, representing the total visual change up to frame t .

To build an intermediate candidate pool of K' non-redundant candidate frames ($K' = \alpha K$ for $\alpha > 1$), we uniformly select frames in the cumulative-variation domain rather than in time. Concretely, we partition $[0, I(T)]$ into K' equal intervals and place a selection target at the midpoint of each:

$$\tau_k = \left(k + \frac{1}{2}\right) \frac{I(T)}{K'}, \quad k = 0, 1, \dots, K' - 1. \quad (1)$$

For each target τ_k , we select the earliest frame t such that $I(t) \geq \tau_k$. Because the targets are equally spaced in visual

variation, more frames are drawn from dynamic segments and fewer from static ones.

Although this step efficiently suppresses temporal redundancy, it does not guarantee visual diversity, i.e., frames at distant timestamps may still be visually similar.

4.2. Feature-Space Entropy Maximization

To remove this residual redundancy, SHIFT refines the temporally selected candidates in the visual embedding space using a log-determinant subset selection criterion. This formulation is based on the objective used in determinantal point processes (DPPs) [13], due to their ability to select a diverse subset from a finite set and naturally model repulsion between similar items. In our setting, this favors subsets that capture complementary visual content while suppressing redundant frames.

Each candidate frame is encoded by a pre-trained visual backbone into a feature embedding, and pairwise cosine similarities are used to build a positive semi-definite kernel matrix $\mathbf{L}$. Under a Gaussian approximation of the feature distribution, the differential entropy of a subset is proportional, up to an additive constant, to the log-determinant of its covariance matrix [12]. Hence, for a subset S , the principal submatrix $\mathbf{L}_S$ summarizes the similarities among the selected frames, and its log-determinant favors subsets with less redundant embeddings, assigning lower values to highly similar selections and higher values to more diverse ones.

The final subset is obtained by solving

$$S^* = \arg \max_{|S|=K} \log \det(\mathbf{L}_S + \lambda \mathbf{I}), \quad (2)$$

where $\mathbf{I}$ is the identity matrix and $\lambda > 0$ is a small regularization constant for numerical stability. This objective assigns higher scores to subsets whose selected frames are less redundant in the embedding space.

While exact optimization is NP-hard [10], the log-determinant objective is monotone submodular, admitting a greedy algorithm with the standard $(1 - 1/e)$ approximation guarantee [16]. We therefore optimize Eq. 2 using the greedy procedure of Chen et al. [3], which iteratively selects the candidate frame with the largest marginal gain in entropy until the budget K is reached. This implementation runs in $\mathcal{O}(K'KD)$ time and $\mathcal{O}(K'(K + D))$ memory, where D is the embedding dimensionality.

The selected frames S^* are then passed to an annotator to obtain bounding-box labels, after which they serve as the sole training set for the target object detector.

5. Experiments

5.1. Experimental Setup

All experiments are conducted on AVADiP-DFS. Each selection method produces a training subset at the target budget, from which a YOLO11X detector [8] is fine-tuned us-

ing COCO-pretrained weights. To isolate the effect of data selection from optimization, all models are trained for the same total number of gradient updates regardless of subset size, with all hyperparameters fixed across methods and budgets; the best-performing checkpoint on the validation set is evaluated on the test set. Methods relying on visual feature extraction use an ImageNet-pretrained ResNet-50 backbone. All experiments were run on a machine with an AMD Ryzen 9 CPU, 186 GB RAM, and a single NVIDIA RTX 5090 GPU.

We compare SHIFT against two baselines: Uniform Frame Sampling (UFS), which selects frames at fixed temporal intervals, and Random Frame Sampling (RFS), which draws frames at random without replacement, averaged over five independent runs. We further evaluate against three Adaptive Frame Sampling (AFS) [26] variants, based on optical flow (OFVD), SSIM (SSVD), or ResNet-50 feature similarity (FSD); and CSOD [14], a coreset method for object detection. Since CSOD requires full annotations prior to selection, training split labels are provided during its selection phase.

5.2. Performance Across Annotation Budgets

Table 2 reports mAP for all methods across seven annotation budgets, ranging from 0.3 fps (1% of the dataset) to 10 fps (33.3%). For comparison, training on the full dataset, without any frame selection, yields 25.3 mAP, serving as a reference for the annotation-efficient methods.

Table 2. Object detection performance (mAP@0.5:0.95) across annotation budgets. Budgets are expressed as sampling rate (fps) and percentage of the full dataset. Training on the full dataset yields 25.3 mAP.

Method	Annotation Budget						
	0.3 1.0%	0.5 1.7%	1 3.3%	2 6.7%	3 10.0%	5 16.7%	10 33.3%
RFS	19.8	19.9	21.8	22.2	24.4	25.4	25.4
UFS	20.8	21.3	21.7	23.7	24.9	25.4	26.3
AFS-FSD	17.2	19.3	20.9	21.6	27.2	25.5	25.2
AFS-OFVD	18.7	18.6	17.5	19.2	24.6	25.2	24.3
AFS-SSVD	17.8	18.4	19.1	20.4	27.1	22.6	24.9
CSOD	18.6	20.1	20.3	21.9	24.8	27.0	24.7
SHIFT (ours)	28.9	29.1	31.3	30.9	31.6	31.8	31.9

Among the baselines, UFS and RFS produce comparable results across all budgets, consistent with Yoon et al. [26], indicating that simple selection strategies are difficult to beat without a principled diversity criterion. As the budget grows and sampled frames become denser, the absence of such a criterion matters more, yet all competing approaches plateau early. Notably, both methods match or exceed the full-dataset reference at 16.7% and 33.3% of the data, suggesting that even trivial subsampling can avoid the redun-

dancy penalty of exhaustive training, an effect that sparsely annotated datasets cannot expose for frame selection evaluation.

The AFS [26] variants underperform UFS and RFS at low budgets, as weighted sampling fails to cover diverse frames under highly constrained budgets. At moderate budgets, AFS-FSD and AFS-SSVD become competitive, peaking at 3 fps (27.2 and 27.1 mAP respectively), briefly surpassing the full-dataset reference, but degrade at higher budgets as over-concentration on high-variance regions introduces redundancy. The optical flow variant underperforms throughout, confirming it is insufficient as a selection criterion without global diversity enforcement.

CSOD [14], despite being designed for static image core-set, is one of the strongest competing methods, reaching 27.0 mAP at 16.7% of the data, yet requires full bounding-box annotations, a constraint that SHIFT eliminates entirely. Although simpler methods such as UFS remain practically valuable, reaching full-dataset performance with one-sixth of data, no competing method delivers consistent gains across the full budget range. This gap is precisely what SHIFT addresses.

SHIFT consistently outperforms all methods across every budget by a substantial margin, demonstrating that principled diversity-driven frame selection is the critical factor in annotation-efficient detector training. Remarkably, with only 1% of the data, SHIFT achieves 28.9 mAP, already surpassing both the full-dataset reference (25.3 mAP) and every competing method at any budget. This suggests that a small but well-chosen set of frames captures sufficient visual diversity for effective training. SHIFT’s performance plateaus around 10% of the dataset (31.6 mAP), with gains under 0.3 mAP at higher budgets, indicating that the method effectively saturates the available visual diversity at a fraction of the annotation cost. Crucially, this behavior is not exclusive to SHIFT: every evaluated method surpasses the full-dataset reference at some budget, confirming that exhaustive annotation is actively detrimental [20, 26].

5.3. Computational Overhead of Frame Selection

While SHIFT delivers superior detection performance, it incurs a higher selection cost than the simplest baselines. Figure 5 reports the total selection time over the training set for each method across budgets; UFS and RFS are omitted as their overhead is negligible.

The AFS variants are budget-independent, as their frame scoring is computed over all frames regardless of the target budget, with fixed costs ranging from 487 s (AFS-SSVD) to 673 s (AFS-OFVD). SHIFT grows moderately with budget, from 535 s at 1.7% to 1089 s at 33%; the fixed SSIM computation (487 s) dominates at low budgets, while embedding extraction becomes the bottleneck at higher budgets. CSOD scales far more aggressively, growing from 878 s to 2175 s

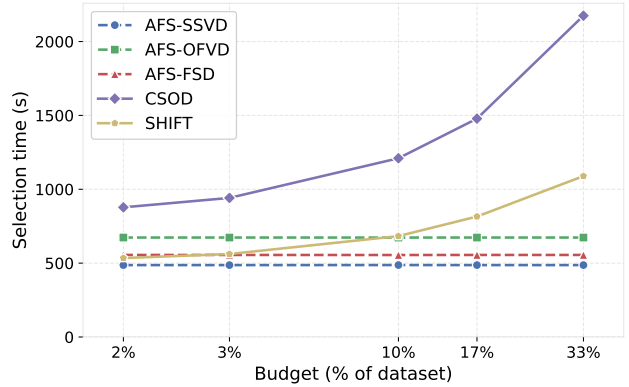


Figure 5. Selection time (seconds) as a function of annotation budget. Dashed lines indicate budget-independent methods, whose cost is determined solely by video length. SHIFT and CSOD scale with budget due to their feature extraction and greedy selection steps.

due to its annotation-dependent greedy selection. Critically, at the lowest budgets, where SHIFT already delivers larger gains over all contending methods, its overhead is comparable to or lower than that of AFS-OFVD and AFS-FSD, making the additional cost well justified relative to the performance improvement achieved.

5.4. Ablation Study

To assess the contribution of each component of SHIFT, we conduct an ablation study at 1 FPS, isolating the effect of the temporal allocation signal and the entropy maximization. Results are reported in Table 3.

Table 3. Ablation study on SHIFT components (mAP at 1 FPS).

Variant	mAP
OFVD temporal allocation	30.1
Pixel difference temporal allocation	30.4
Uniform temporal allocation	30.9
W/o entropy maximization	29.3
Full SHIFT	31.3

We first evaluate alternative variation signals for temporal allocation: replacing SSIM with optical flow (OFVD) or frame pixel difference degrades performance by 1.2 and 0.9 mAP, respectively, confirming that SSIM is more effective and robust structural signal for capturing training-relevant visual change. Next, replacing Stage 1 entirely with uniform sampling yields only a marginal drop (0.4 mAP), suggesting that reducing temporal redundancy in the candidate pool meaningfully supports the diversity refinement which follows. Finally, removing the diversity refinement step entirely produces the largest drop (2.0 mAP), establishing entropy maximization as the critical component of SHIFT.

Together, these results confirm the complementary de-



Figure 6. Qualitative comparison between frames selected using UFS, AFS-SSVD and SHIFT.

sign of SHIFT: reducing temporal redundancy provides a diverse candidate pool, which entropy maximization then refines into the most informative subset.

5.5. Qualitative Analysis

Figure 6 provides a qualitative comparison between frames selected by UFS, AFS-SSVD and the proposed SHIFT at a budget of 0.5,fps (1.7% of frames) on a randomly selected video. While quantitative results already demonstrate the performance gain of SHIFT, this example highlights complementary behaviors of different frame selection strategies. In particular, it illustrates how the absence of diversity enforcement leads to either missed events or redundant selections, regardless of the sampling strategy.

By definition, UFS selects frames at uniform time intervals, ignoring the fact that informative non-redundant events might not be evenly distributed over time, which is particularly relevant in dynamic driving videos. In Figure 6, a motorcycle crosses the video between frames with index 80 and 150. Because UFS samples every 60 frames, the event is captured only at a distant perspective. At the same time, part of the frame budget is spent on frames with similar visual content, such as frames 0 and 60.

On the other hand, AFS-SSVD exhibits a different limitation by selecting a highly concentrated subset, spanning only 22 frames. As a result, all of the selected images correspond to nearly consecutive time steps and are visually very similar, contributing little additional information. This behavior reflects the method’s focus on local variation between frames without explicitly enforcing diversity across the selected subset, which may lead to clusters of near-duplicate samples.

In contrast, SHIFT selects samples according to the rate of visual variation in the video, but also accounts for diversity. Therefore, more frames are allocated to informative segments, allowing the motorcycle crossing event to be better represented when compared to UFS. At the same time, the selected frames are better spread across the video dura-

tion, not choosing a highly localized subset of frames like AFS-SSVD. Together, these observations highlight the importance of simultaneously considering temporal variation and global diversity.

6. Conclusion

In this work, we introduced AVADiP-DFS, the first benchmark with exhaustive 30 fps per-frame bounding-box annotations designed for evaluating frame selection methods for object detector training on driving video. We also proposed SHIFT, a training-free two-stage method that jointly addresses temporal non-redundancy and semantic diversity through variation-based temporal allocation and greedy entropy maximization over visual embeddings. An exploratory analysis of AVADiP-DFS revealed that driving video exhibits high local frame similarity and persistent semantic redundancy across temporal scales, validating the design choices of SHIFT.

Our experiments demonstrated that SHIFT consistently surpasses full-dataset performance with only 1% of frames, confirming that temporal redundancy is a bottleneck in detector training from driving video. While evaluated exclusively on driving video, SHIFT is broadly applicable to any sequential data domain with a temporal similarity signal and a feature embedding space. In future work, we will extend SHIFT to video understanding tasks such as action recognition, and explore learned temporal allocation signals as task-aware alternatives to the fixed SSIM-based variation measure.

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